ROBOT CAR PROJECT

MODULE 04

04

Software & App Guide

Install the tools, upload both boards, connect the phone and master every control.



Name _____ Class / Group _____ Date _____

4WD · Arduino UNO · ESP32-CAM
Edition 2026

MODULE 04

1 Software & uploading

The car runs two programs: one on the **Arduino UNO** (motors, sensor, alarm) and one on the **ESP32-CAM** (WiFi, camera, phone app). Both are uploaded with the free **Arduino IDE 2**. No extra libraries are needed.

LEARNING OBJECTIVES · ABOUT 60 MIN

By the end of this module you can:

- Install the Arduino IDE and the ESP32 board package
- Upload code to an Arduino UNO and to an ESP32-CAM
- Read the Serial Monitor to check that a board works
- Connect a phone and use every feature of the app

1 • Install the software

1. Download and install **Arduino IDE 2** from `arduino.cc/en/software`.
2. Open **File → Preferences**. In *Additional boards manager URLs* paste:

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```

3. Open **Tools → Board → Boards Manager**, search **esp32** and install **"esp32 by Espressif Systems"** (version 3.x).
4. Open the project folder. The code is in `firmware/uno_controller` and `firmware/esp32cam_robot`.

2 • Board settings

Setting	Arduino UNO	ESP32-CAM
Sketch	`uno_controller.ino`	`esp32cam_robot.ino`
Tools → Board	Arduino AVR Boards → **Arduino Uno**	esp32 → **AI Thinker ESP32-CAM**
Partition scheme	–	**Huge APP (3MB No OTA)**
PSRAM	–	Enabled (if the menu exists)
Port	the UNO's USB port	the MB board's port (CH340)

Arduino Nano or a clone?
Choose **Arduino Nano** and, if the upload fails, set **Processor → ATmega328P (Old Bootloader)**.

3 • Upload the Arduino UNO

1. Open `uno_controller.ino` and select the board and port.
2. Click **Upload** (→). You can leave all wires connected.
3. After a reset you hear a **short beep** and the **headlights blink**: the UNO is running.

4 • Upload the ESP32-CAM

Option A: ESP32-CAM-MB board EASY

1. Unplug the ESP32-CAM from the car and press it onto the MB board.
2. Connect the USB cable, select the port, click **Upload**.
3. Stuck at `Connecting....`? Hold **IO0**, tap **RST**, release IO0.
4. When it is done, press **RST** to start the program.

Option B: FTDI adapter

FTDI	ESP32-CAM
5V (jumper on 5 V)	5V
GND	GND
TX	U0R
RX	U0T
–	**IO0** → **GND** while uploading

Press RST, upload, then **remove the IO0-GND wire** and press RST again.

Check the Serial Monitor (115200 baud)

```
=== ROBOT CAR ESP32-CAM ===
WiFi network "RobotCar" / password "12345678"
Open  http://192.168.4.1  on your phone
Ready.
```

"Camera init FAILED"
The ribbon cable is not fully inserted (open the latch, push it in, close it) or the wrong board type is selected.

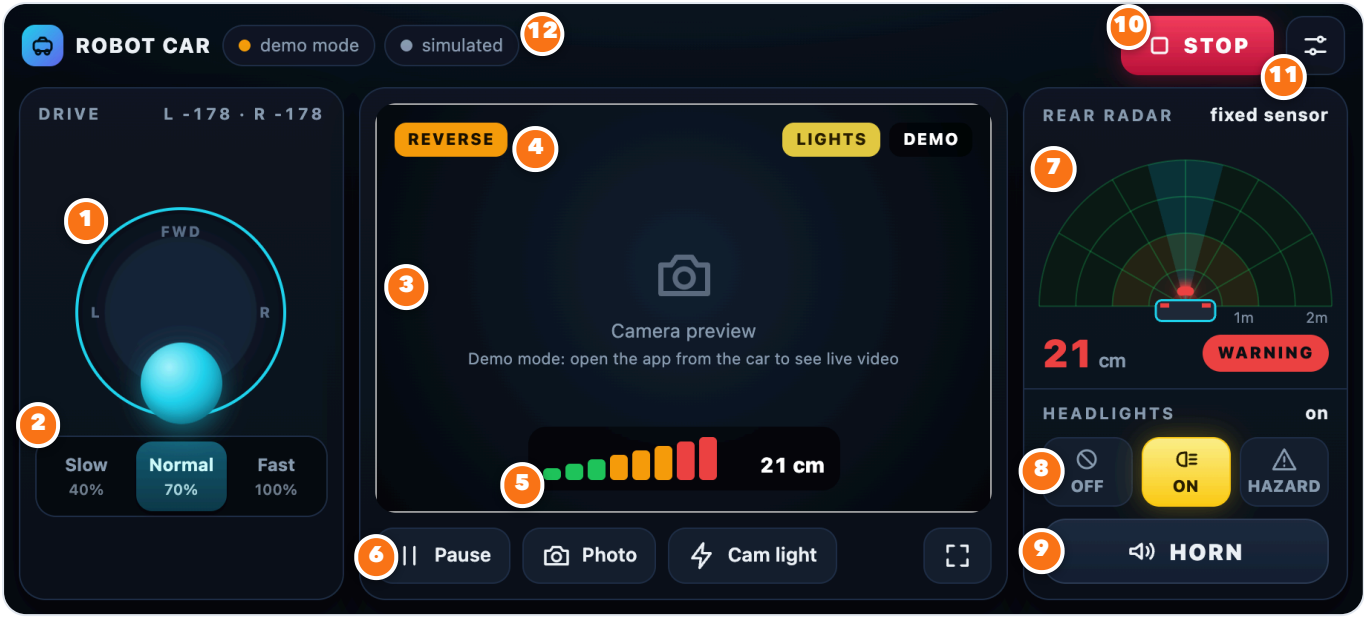
5 • Connect the phone

1. Switch on the car. The small red LED on the back of the ESP32-CAM **blinks**: it is waiting for a phone.
2. On the phone, open WiFi settings → network **RobotCar**, password **12345678**. If the phone warns "no internet", choose **stay connected**.
3. Open the browser at **http://192.168.4.1**. The chips show **connected** and **car ok**.
4. Turn the phone sideways and tap ⛶ for full screen. Tip: **Add to Home screen** so it opens like an app.
5. The first time, a short **tips** screen explains the controls. Tap **Let's drive**.
6. Optional: on a PC joined to the same WiFi, open **http://192.168.4.1/view** for the classroom viewer (section 3).

Built-in safety
Release the joystick or press the red **STOP**: the car stops. Lock the phone, switch app or lose WiFi: it stops within 0.6 s. It never reverses closer than the auto-brake distance (forward still works).
Keyboard (app on a laptop): W A S D or arrows = drive · Space = horn · L = headlights · Esc = stop.

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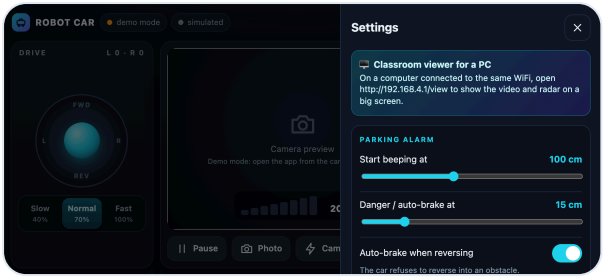
2 Using the app



The phone app in landscape: reversing towards an obstacle at 21 cm, headlights on

#	Element	What it does
1	**Joystick**	Drag up/down = forward/back, left/right = turn. Release = stop. (A D-pad mode is in the settings.)
2	Speed gears	**Slow** 40 % · **Normal** 70 % · **Fast** 100 %. Start with Slow
3	**Live video**	From the front camera. A flashing **red border** means an obstacle is close behind while reversing
4	Status tags	**REVERSE** while backing up, **AUTO-BRAKE** when the car refuses to go back, **LIGHTS** when the headlights are on
5	Parking bar	More bars lit = closer, green → amber → red. Shows the distance in cm
6	Video buttons	**Pause** the video · save a **Photo** on the phone · **Cam light** = the camera's own flash LED · ⛶ full screen
7	**Rear radar**	Distance behind the car: green, amber or red zone, beam cone, big distance number and zone name
8	**Headlights**	**OFF** · **ON** · **HAZARD** (headlights and red rear LED blink together)
9	**Horn**	Hold to sound the car's buzzer
10	**STOP**	Stops the car at once
11	Settings	See the next page
12	Status chips	**connected** to the car and **car ok** (the UNO answers). Amber or red = check the WiFi

Settings



Settings sheet (tap the sliders icon)



Portrait mode works too

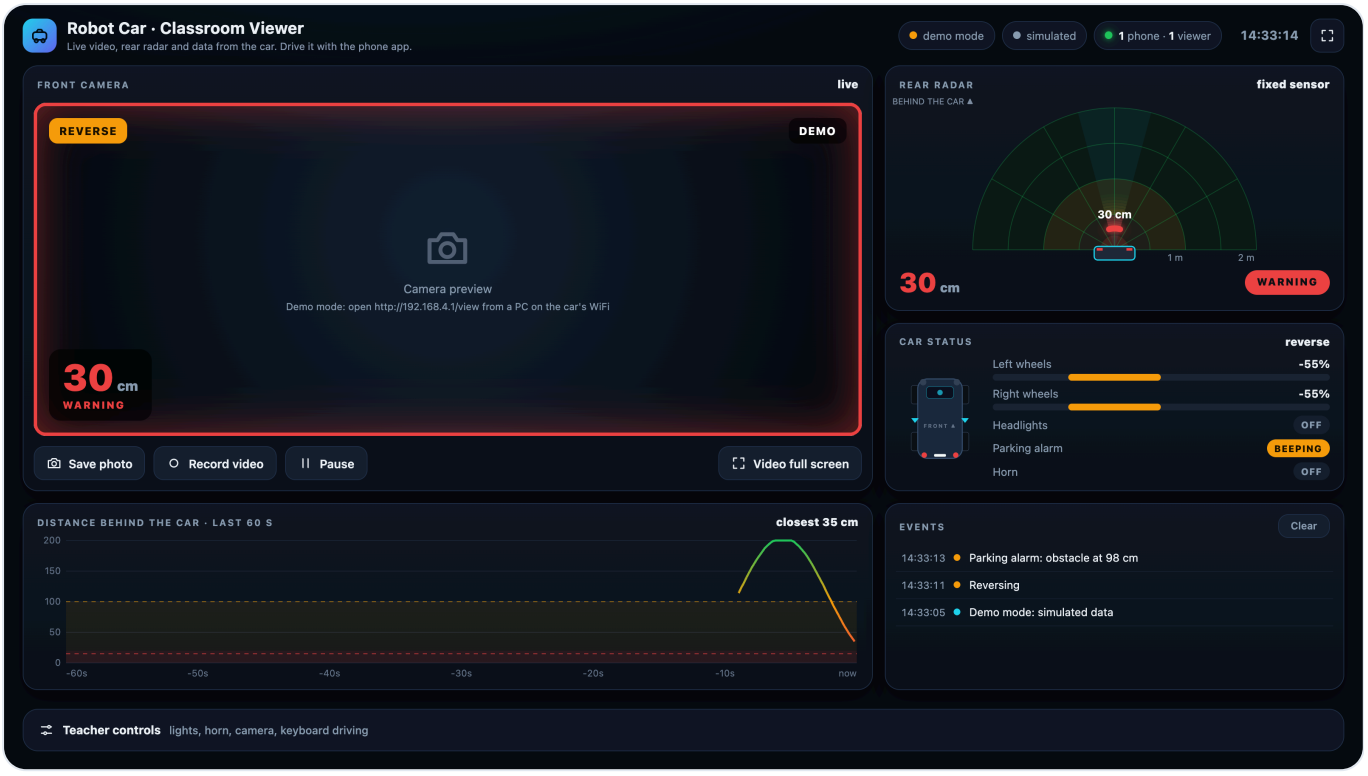
Setting	Meaning
Parking alarm	
Start beeping at	Beeping starts below this distance (default 100 cm)
Danger / auto-brake at	Continuous tone + brake below this (default 15 cm)
Auto-brake when reversing	Block reversing near obstacles
Alarm also when not reversing	Beep in any direction
Mute the car buzzer	The LED still flashes
Beep on this phone / Vibrate	The phone copies the alarm
Driving	
Controls	Joystick or D-pad
Exact speed	Fine-tune the speed gear (30–100 %)
Turn sensitivity	How sharp the car turns
Camera	
Resolution / picture quality	VGA = default. QVGA = smoothest video
Cam light brightness	Brightness of the camera's flash LED
Flip / Mirror	If the image is upside down or reversed
Radar & connection	
Radar range	Distance shown at the edge of the radar
Car address	Leave 192.168.4.1 unless you changed the WiFi mode
Show the tips again	Opens the first-time tips screen

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3 Classroom viewer on a PC

The car serves a **second page** made for a computer or a projector. One student drives with the phone while the class watches the video, the radar and live data on the big screen. Nothing needs to be installed.

1. Connect the PC to the WiFi **RobotCar** (password **12345678**).
2. Open **http://192.168.4.1/view** in Chrome, Edge or Firefox. (The phone app shows the same link in its settings, and a banner on a PC.)
3. Press **F11** or the full-screen button for the projector.



The classroom viewer (here in demo mode, reversing at 28 cm)

Part of the page	What it shows or does
Front camera	Big live video with the REVERSE / AUTO-BRAKE tags and a large distance number while reversing
Save photo · Record video	Photo as JPG with a data strip (time, distance, speeds). Video as a WebM file with the distance drawn on it
Rear radar	The same radar as on the phone, larger
Car status	Picture of the car: headlight beams, rear and reverse lights, wheel arrows, left/right motor bars, alarm and horn
Distance chart	The distance behind the car for the last 60 s, with the warning and danger lines. Closest value shown
Events	Log with time: reversing, alarms, auto-brake, lights, connection
Teacher controls	Headlights, horn, cam light, radar range, and **keyboard driving** (W A S D / arrows, Space = horn, Esc = stop)

How many screens?

Up to **3 video screens** at once (for example the phone and two PCs). All screens get the radar and data. Close unused tabs if a screen says “too many viewers”.

Exam and report ideas

Record a video of a parking manoeuvre, then use the chart to read the closest distance and the moment the auto-brake stopped the car.

Customising the car

File · setting	Default	Change it to...
`config.h` · AP_SSID / AP_PASSWORD	RobotCar / 12345678	give every group its own WiFi name
`config.h` · STA_SSID	empty	join a school WiFi or phone hotspot instead
`config.h` · CAM_VFLIP	0	1 if the camera is mounted upside down
`uno_controller.ino` · MIN_PWM	90	raise if the car doesn't move at low speed
`uno_controller.ino` · INVERT_LEFT/RIGHT	false	flip a side that drives backwards
`uno_controller.ino` · USE_SERVO	0	1 = scanning radar with an SG90 servo on A1

Several cars in one classroom

Give each car a different `AP_SSID` (RobotCar1, RobotCar2 …) and a different `AP_CHANNEL` (1, 6, 11) so they don't interfere.

Editing the pages

The phone app is `web/index.html`, the PC viewer `web/view.html`. Open them on a computer to try them in **demo mode**. Then run `python3 tools/build_ui.py` and upload the ESP32-CAM again.